AI Assistant Coding

Assignment-15

Task 1– Real-Time Application: Online Food Ordering API

Scenario:

Design a simple API for an online food ordering system.

Requirements:

• Endpoints required:

• GET /menu → List available dishes

• POST /order → Place a new order

• GET /order/{id} → Track order status

• PUT /order/{id} → Update an existing order (e.g., change items)

• DELETE /order/{id} → Cancel an order

• **AI should generate:**

• REST API code

• Suggested improvements (like authentication, pagination)

Expected Output:

• Fully working API simulating a food ordering backend.

Prompt:

Create a Node.js Express REST API for an online food ordering system.

Requirements:

- GET /menu → return list of dishes

- POST /order → create new order with items

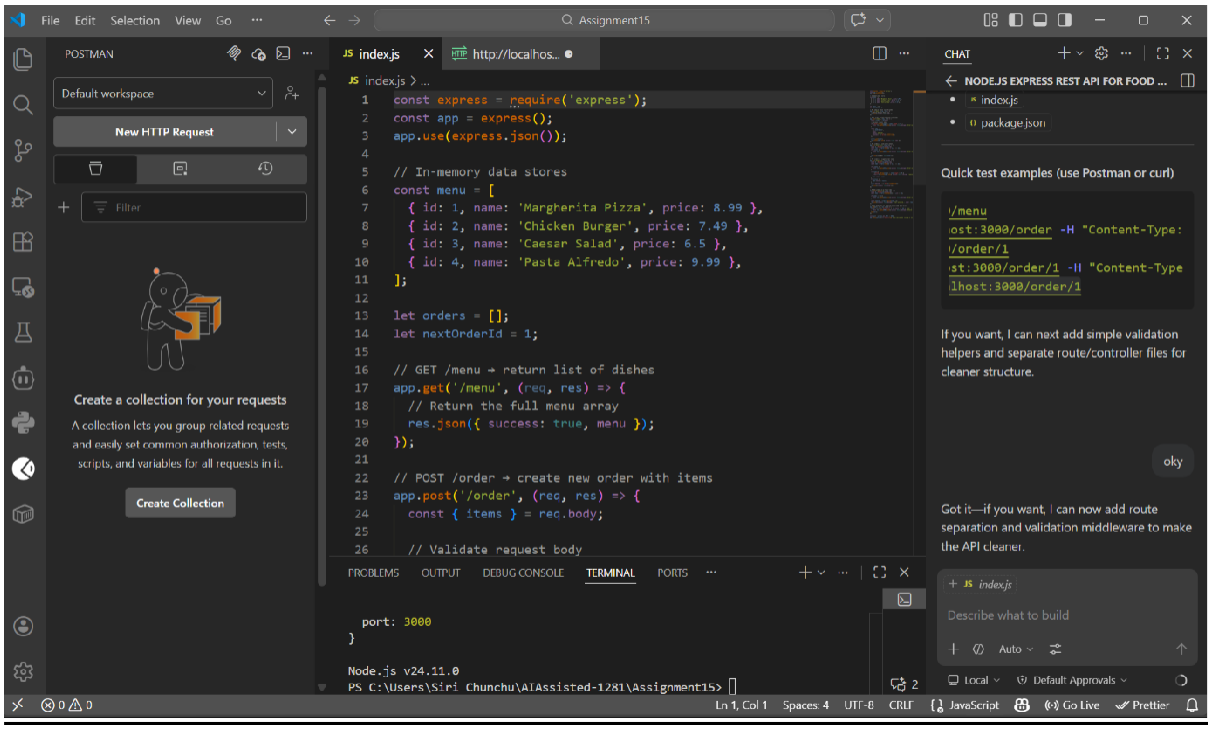
- GET /order/:id → get order status

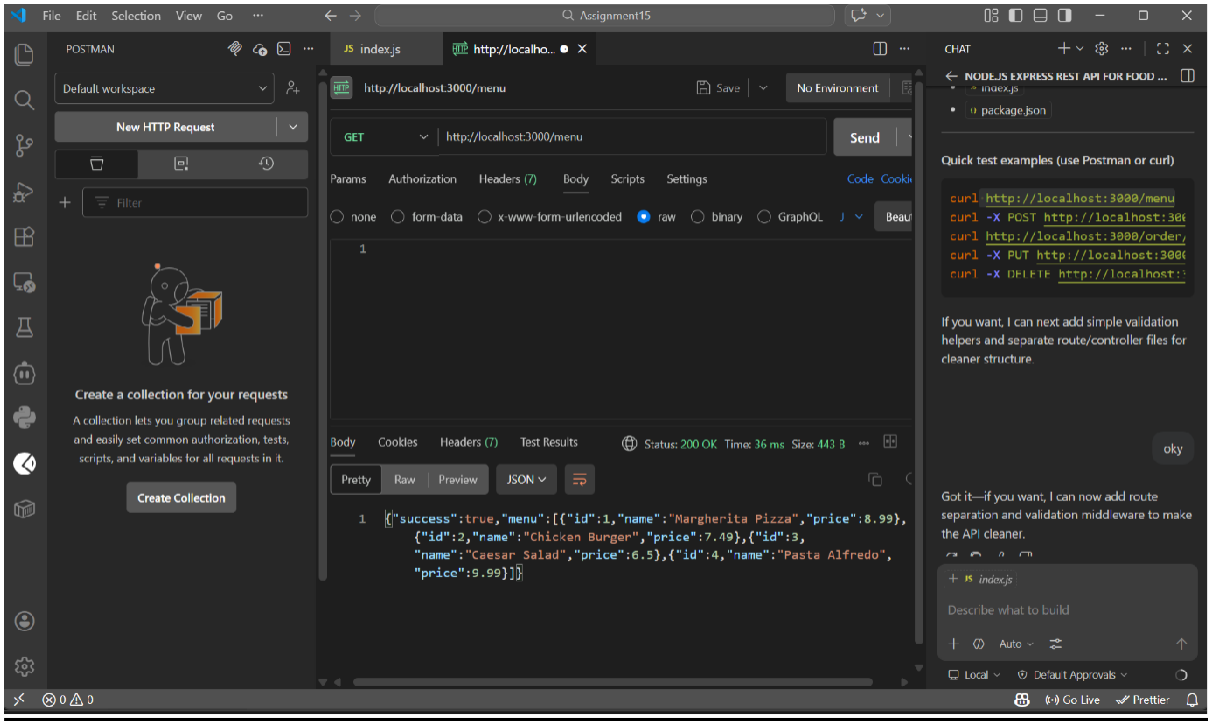
- PUT /order/:id → update order items

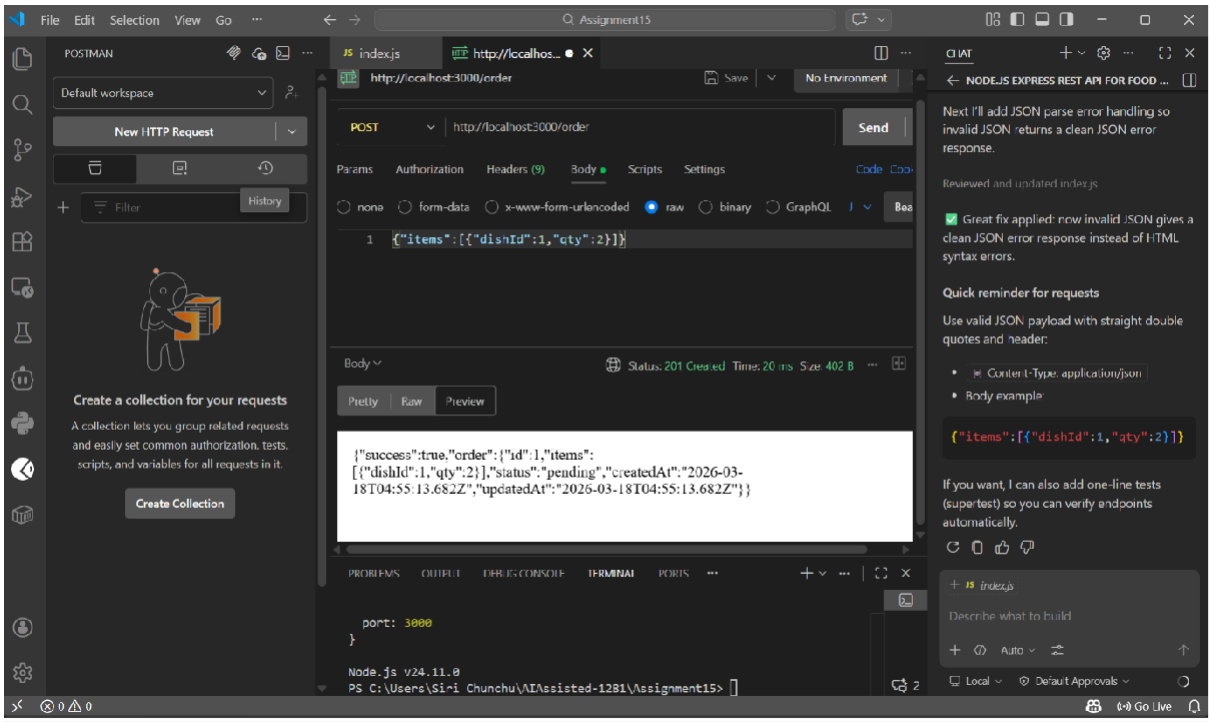
- DELETE /order/:id → cancel order

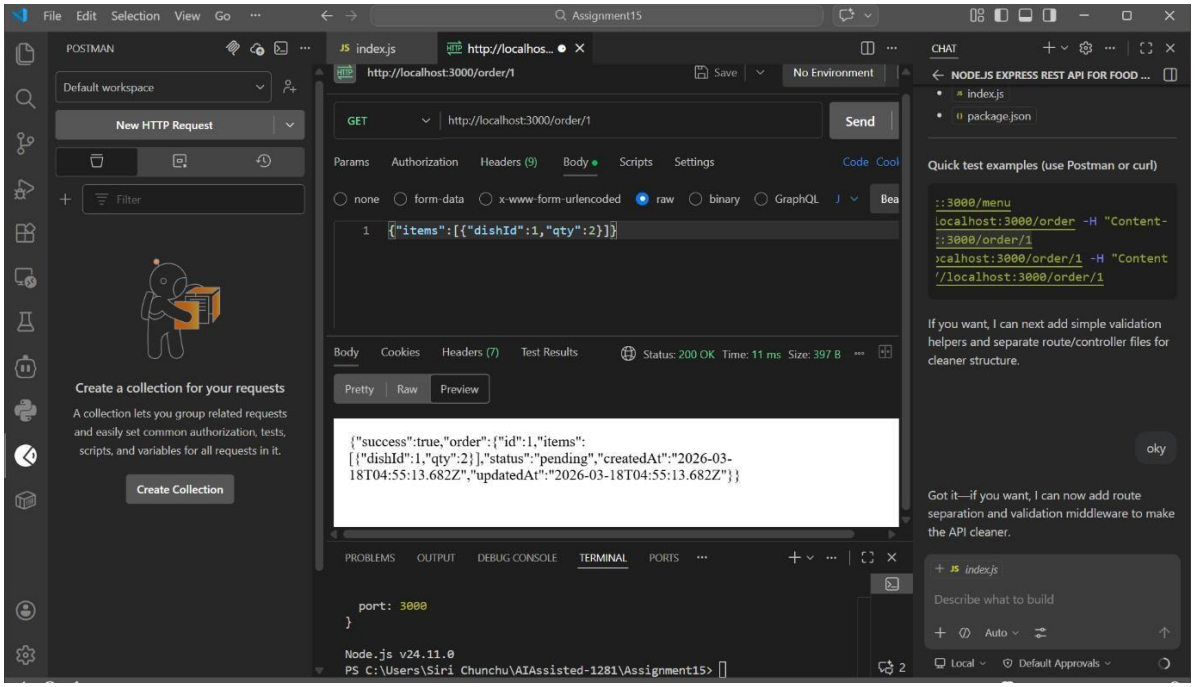
Use in-memory arrays. Add proper comments for each endpoint.

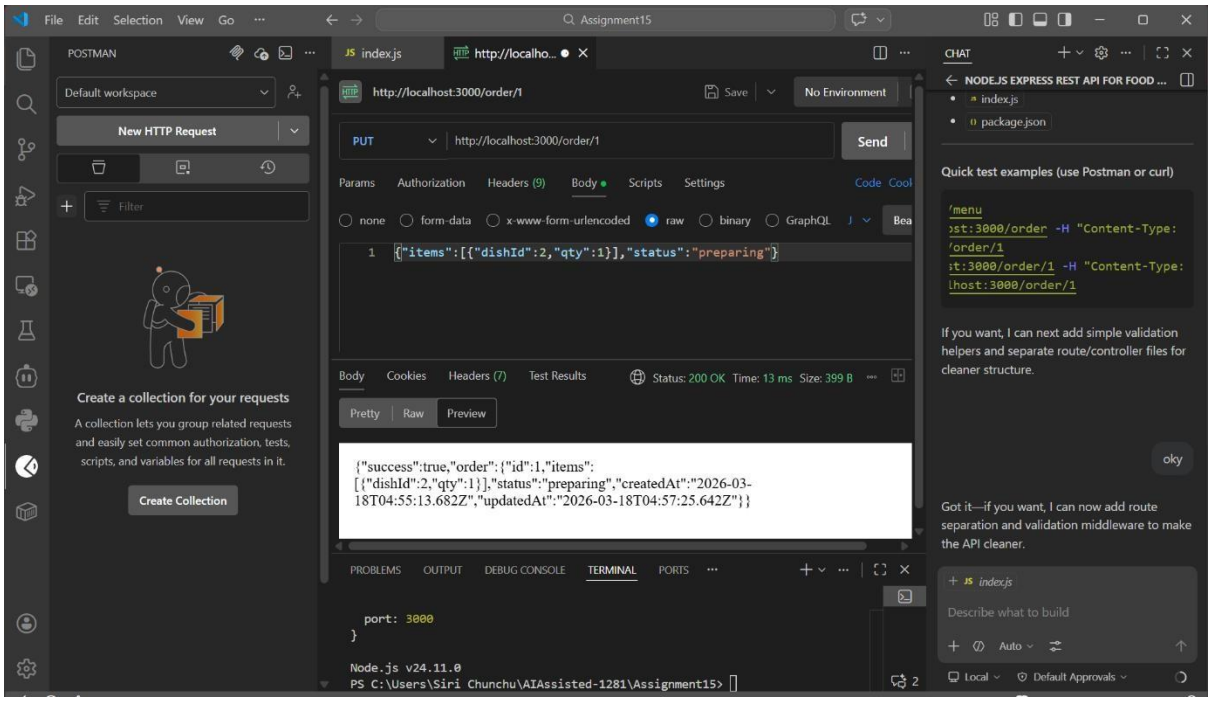
AI Generated:

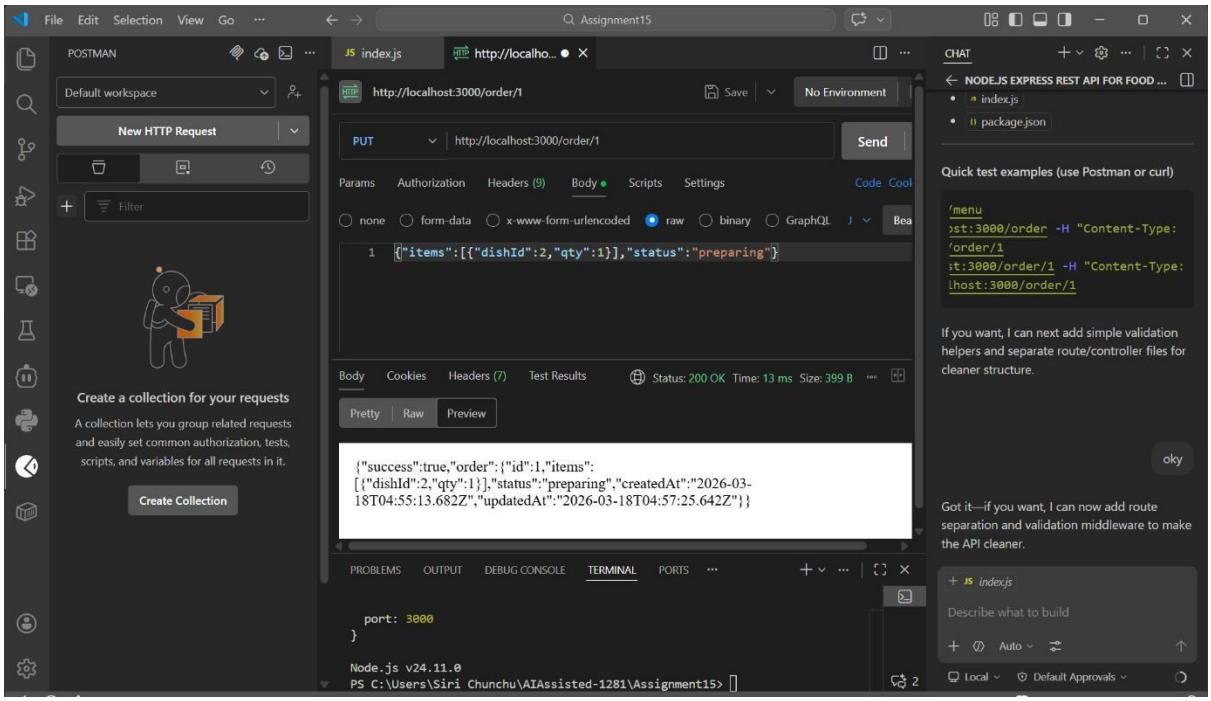





File Edit Selection View Go
Assignment15
POSTMAN
Default workspace
New HTTP Request
History
Filter
Create a collection for your requests
A collection lets you group related requests and easily set common authorization, tests, scripts, and variables for all requests in it.
Create Collection
index.js
http://localhos...
http://localhost:3000/order
Save
No Environment
POST
http://localhost:3000/order
Send
Params Authorization Headers (9) Body Scripts Settings
none form-data x-www-form-urlencoded raw binary GraphQL
{"items":[{"dishId":1,"qty":2}]}
Body
Status: 201 Created Time: 20 ms Size: 402 B
Pretty Raw Preview
{"success":true,"order":{"id":1,"items":[{"dishId":1,"qty":2}],"status":"pending","createdAt":"2026-03-18T04:55:13.682Z","updatedAt":"2026-03-18T04:55:13.682Z"}}
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
port: 3000
}
Node.js v24.11.0
PS C:\Users\Siri Chunchu\AIAssisted-1281\Assignment15>
CHAT
NODE.JS EXPRESS REST API FOR FOOD ...
Next I'll add JSON parse error handling so invalid JSON returns a clean JSON error response.
Reviewed and updated index.js
Great fix applied: now invalid JSON gives a clean JSON error response instead of HTML syntax errors.
Quick reminder for requests
Use valid JSON payload with straight double quotes and header:
Content-Type: application/json
Body example
{"items":[{"dishId":1,"qty":2}]}
If you want, I can also add one-line tests (supertest) so you can verify endpoints automatically.
index.js
Describe what to build
Auto
Local
Default Approvals
Go Live


File Edit Selection View Go
Assignment15
POSTMAN
Default workspace
New HTTP Request
Filter
Create a collection for your requests
A collection lets you group related requests and easily set common authorization, tests, scripts, and variables for all requests in it.
Create Collection
index.js
http://localhos...
http://localhost:3000/order/1
Save
No Environment
GET
http://localhost:3000/order/1
Send
Params Authorization Headers (9) Body Scripts Settings
none form-data x-www-form-urlencoded raw binary GraphQL
{"items":[{"dishId":1,"qty":2}]}
Body Cookies Headers (7) Test Results
Status: 200 OK Time: 11 ms Size: 397 B
Pretty Raw Preview
{"success":true,"order":{"id":1,"items":[{"dishId":1,"qty":2}],"status":"pending","createdAt":"2026-03-18T04:55:13.682Z","updatedAt":"2026-03-18T04:55:13.682Z"}}
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
port: 3000
}
Node.js v24.11.0
PS C:\Users\Siri Chunchu\AIAssisted-1281\Assignment15>
CHAT
NODE.JS EXPRESS REST API FOR FOOD ...
index.js
package.json
Quick test examples (use Postman or curl)
::3000/menu
localhost:3000/order -H "Content-
::3000/order/1
ocalhost:3000/order/1 -H "Content
'/localhost:3000/order/1
If you want, I can next add simple validation helpers and separate route/controller files for cleaner structure.
oky
Got it—if you want, I can now add route separation and validation middleware to make the API cleaner.
index.js
Describe what to build
Auto
Local
Default Approvals





Explanation

This API simulates a basic online food ordering system using REST principles.
The **GET /menu** endpoint retrieves available dishes.
The **POST /order** endpoint allows users to place a new order by sending selected items.
The **GET /order/:id** endpoint helps in tracking the status of an order.
The **PUT /order/:id** endpoint is used to update order details such as items.
The **DELETE /order/:id** endpoint cancels an existing order.
All data is stored in an in-memory array, making it simple and fast for testing.

Task 2 – Employee Payroll API

Task:

Create an API for managing employees and their salaries.

Instructions:

• **Endpoints required:**

• GET /employees → List all employees

• POST /employees → Add a new employee with salary details

• PUT /employees/{id}/salary → Update salary of an employee

• DELETE /employees/{id} → Remove employee from system

• **Use AI to:**

• Suggest data model structure.

• Add comments/docstrings for all endpoints.

Expected Output:

• Payroll management API with validation

• Secure and structured data handling

• Clear API documentation

<u>Prompt:</u>

Create a Node.js Express REST API to manage employees and salaries.

Requirements:

- GET /employees → list all employees

- POST /employees → add employee with name, role, salary

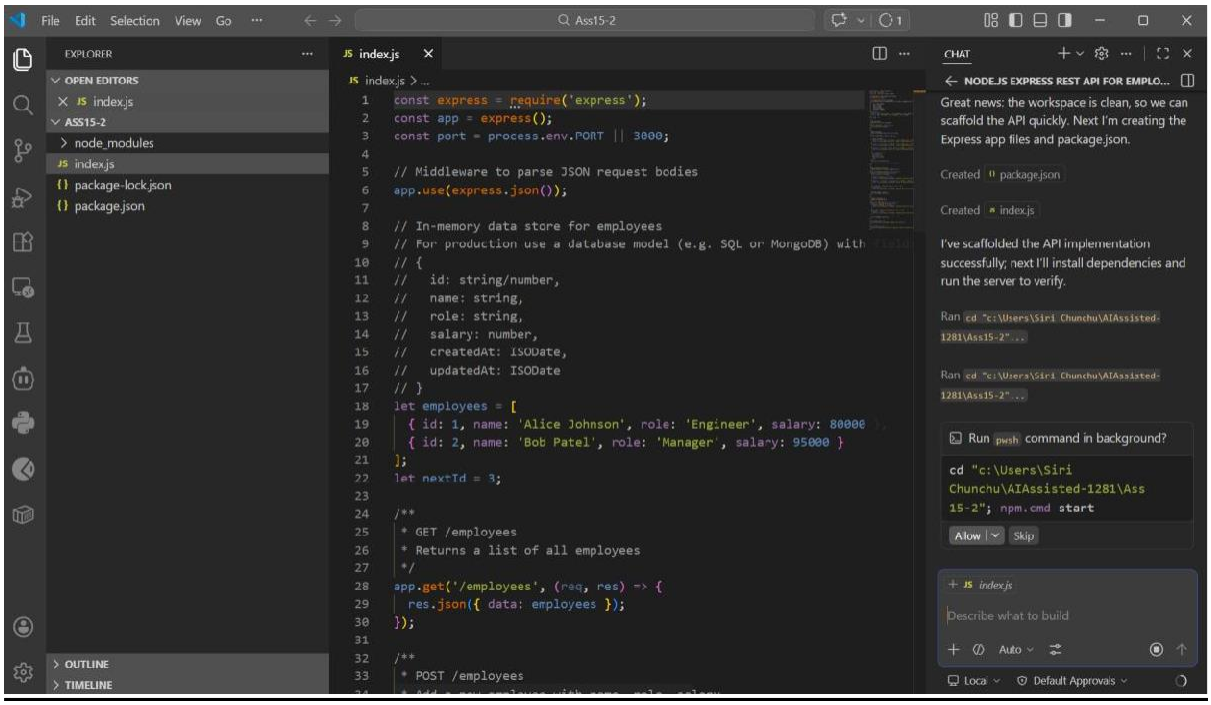
- PUT /employees/:id/salary → update salary

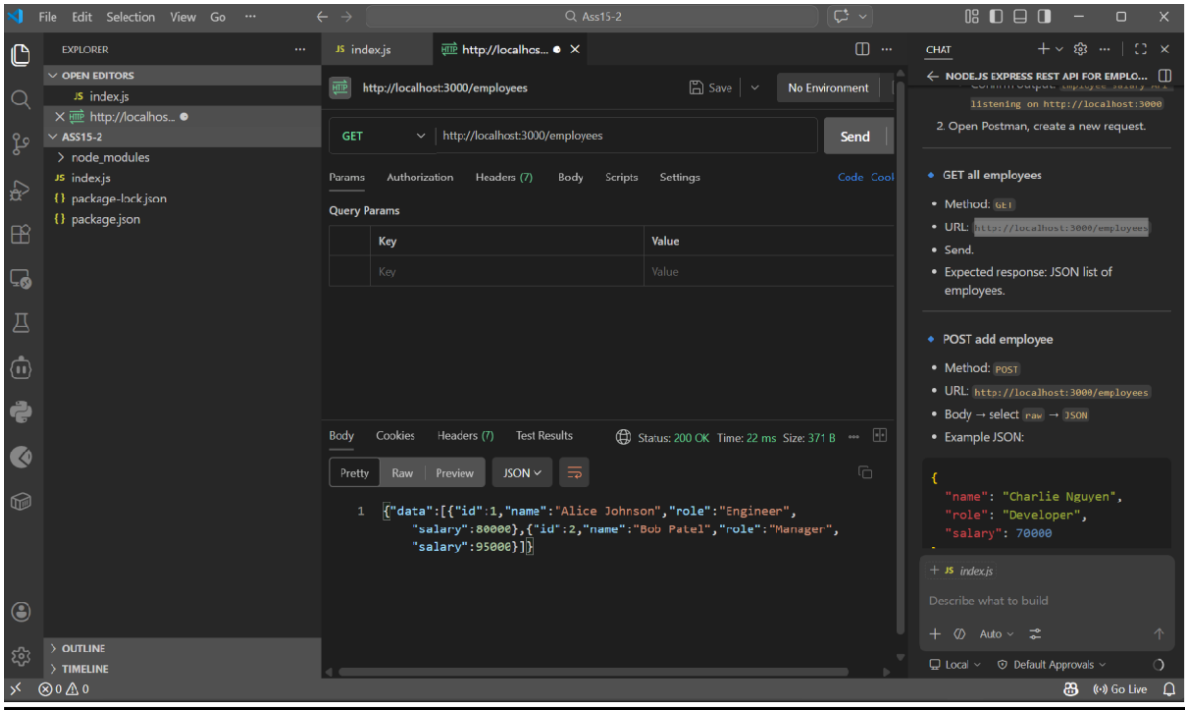
- DELETE /employees/:id → remove employee

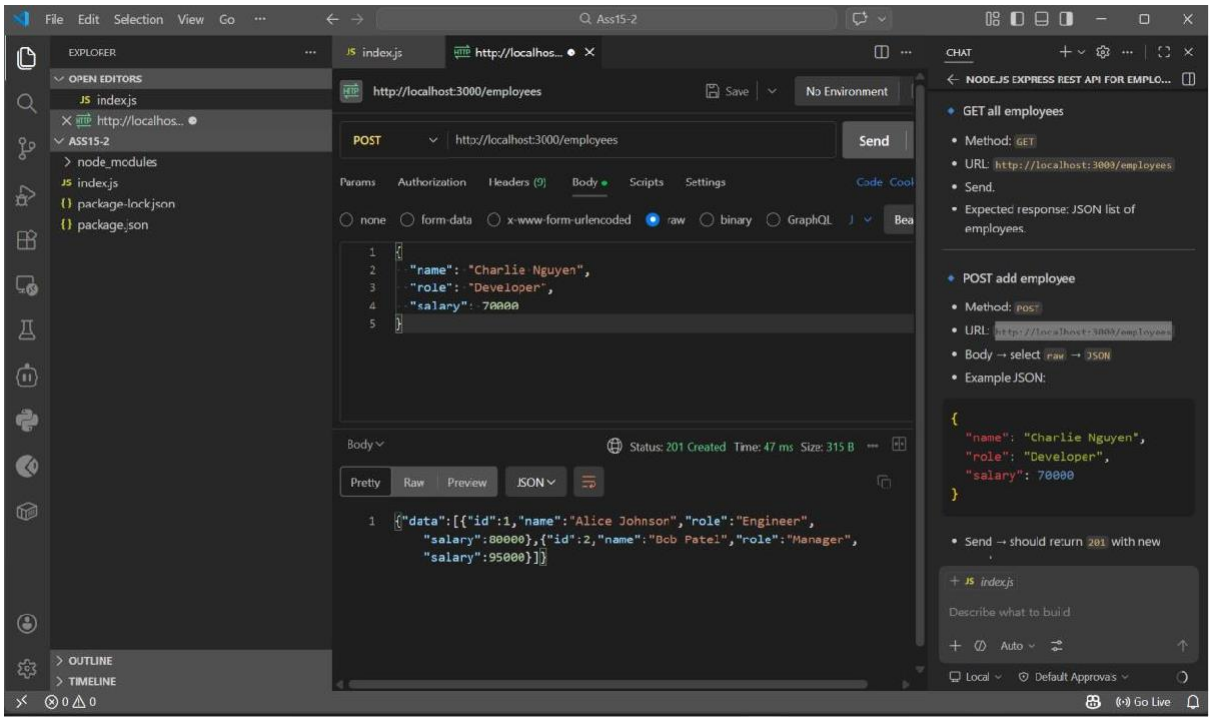
Add validation and comments for each endpoint.

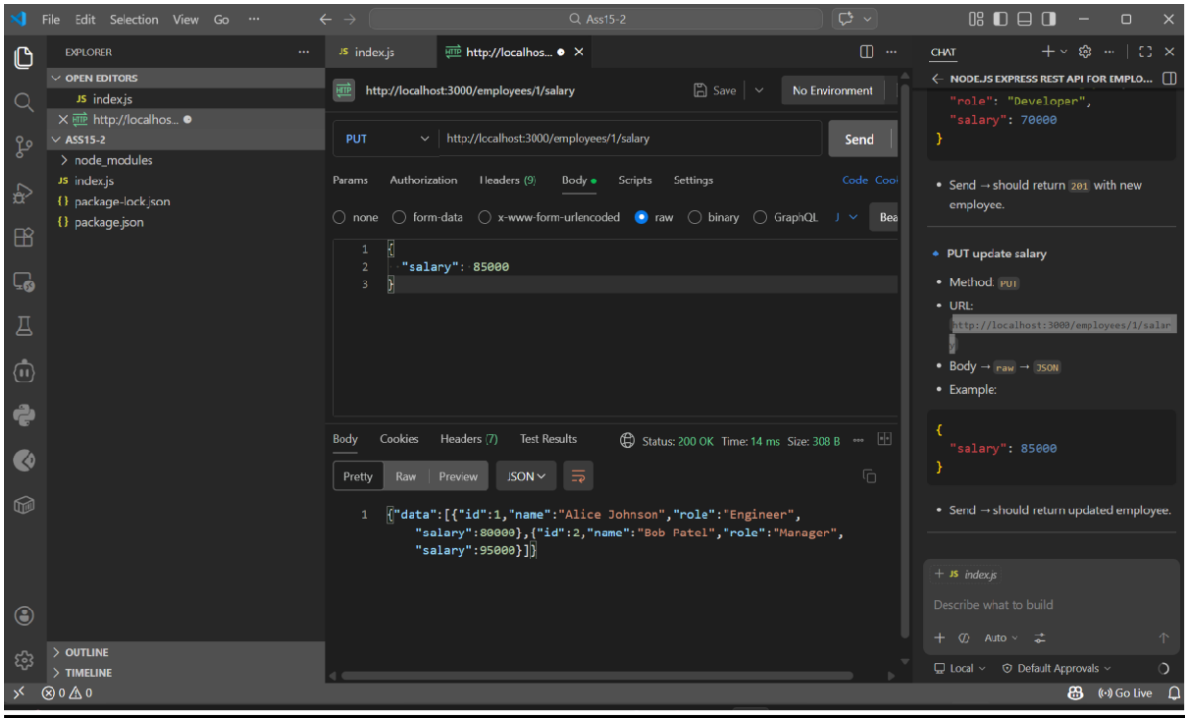
Also suggest a proper data model.

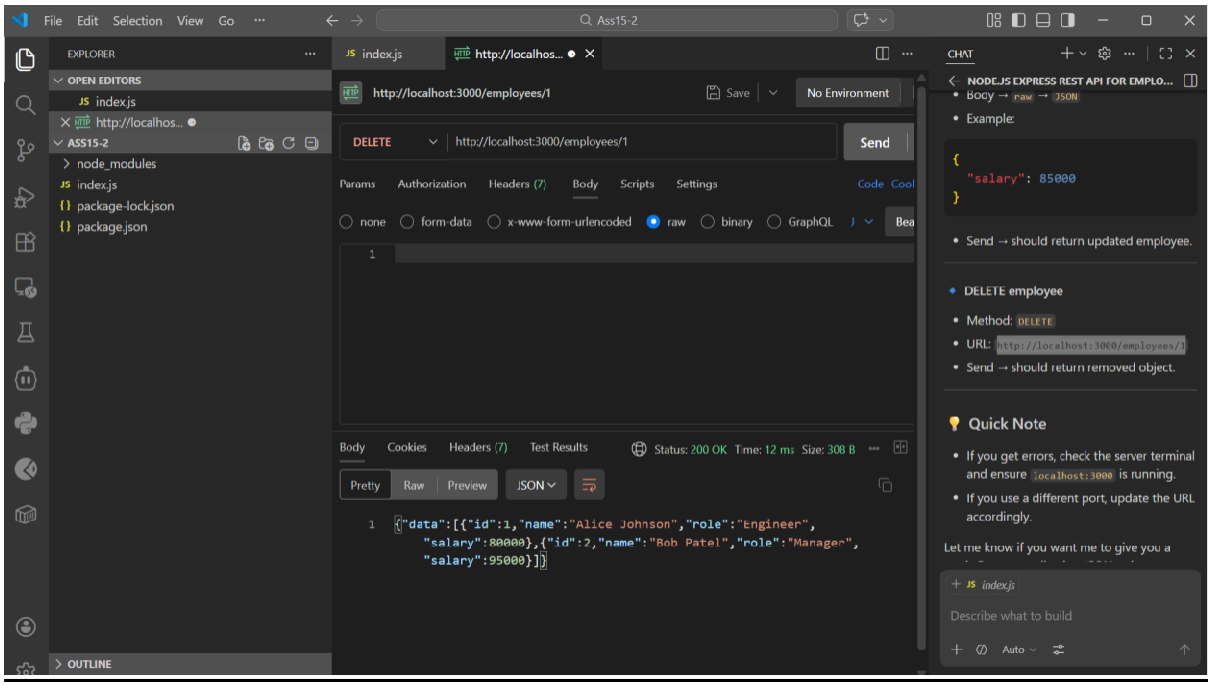
<u>AI Generated:</u>


File Edit Selection View Go
Ass15-2
EXPLORER
OPEN EDITORS
index.js
ASS15-2
node_modules
index.js
package-lock.json
package.json
OUTLINE
TIMELINE
index.js > ...
const express = require('express');
const app = express();
const port = process.env.PORT || 3000;
// Middleware to parse JSON request bodies
app.use(express.json());
// In-memory data store for employees
// For production use a database model (e.g. SQL or MongoDB) with
// {
// id: string/number,
// name: string,
// role: string,
// salary: number,
// createdAt: ISODate,
// updatedAt: ISODate
// }
let employees = [
{ id: 1, name: 'Alice Johnson', role: 'Engineer', salary: 80000 },
{ id: 2, name: 'Bob Patel', role: 'Manager', salary: 95000 }
];
/**
* GET /employees
* Returns a list of all employees
*/
app.get('/employees', (req, res) => {
res.json({ data: employees });
});
/**
* POST /employees
CHAT
NODE.JS EXPRESS REST API FOR EMPLO...
Great news: the workspace is clean, so we can scaffold the API quickly. Next I'm creating the Express app files and package.json.
Created package.json
Created index.js
I've scaffolded the API implementation successfully; next I'll install dependencies and run the server to verify.
Ran cd "c:\Users\Siri Chunchu\AIAssisted-1281\Ass15-2"...
Ran cd "c:\Users\Siri Chunchu\AIAssisted-1281\Ass15-2"...
Run pwsh command in background?
cd "c:\Users\Siri Chunchu\AIAssisted-1281\Ass15-2"; npm.cmd start
Allow
Skip
index.js
Describe what to build
Auto
Local
Default Approvals


File Edit Selection View Go
Ass15-2
EXPLORER
OPEN EDITORS
index.js
http://localhos...
ASS15-2
node_modules
index.js
package-lock.json
package.json
OUTLINE
TIMELINE
index.js
http://localhos...
http://localhost:3000/employees
Save
No Environment
GET
http://localhost:3000/employees
Send
Params
Authorization
Headers (7)
Body
Scripts
Settings
Code
Cool
Query Params
Key
Value
Key
Value
Body
Cookies
Headers (7)
Test Results
Status: 200 OK
Time: 22 ms
Size: 371 B
Pretty
Raw
Preview
JSON
{"data":[{"id":1,"name":"Alice Johnson","role":"Engineer",
"salary":80000},{"id":2,"name":"Bob Patel","role":"Manager",
"salary":95000}]}
CHAT
NODE.JS EXPRESS REST API FOR EMPLO...
listening on http://localhost:3000
2. Open Postman, create a new request.
GET all employees
Method: GET
URL: http://localhost:3000/employees
Send.
Expected response: JSON list of employees.
POST add employee
Method: POST
URL: http://localhost:3000/employees
Body → select raw → JSON
Example JSON:
{
"name": "Charlie Nguyen",
"role": "Developer",
"salary": 70000
index.js
Describe what to build
Auto
Local
Default Approvals
Go Live


File Edit Selection View Go
Ass15-2
EXPLORER
OPEN EDITORS
index.js
http://localhos...
ASS15-2
node_modules
index.js
package-lock.json
package.json
OUTLINE
TIMELINE
http://localhost:3000/employees
Save
No Environment
POST
http://localhost:3000/employees
Send
Params Authorization Headers (9) Body Scripts Settings
none form-data x-www-form-urlencoded raw binary GraphQL
{
"name": "Charlie Nguyen",
"role": "Developer",
"salary": 70000
}
Body
Status: 201 Created Time: 47 ms Size: 315 B
Pretty Raw Preview JSON
{"data":[{"id":1,"name":"Alice Johnson","role":"Engineer",
"salary":80000},{"id":2,"name":"Bob Patel","role":"Manager",
"salary":95000}]}
CHAT
NODE.JS EXPRESS REST API FOR EMPLO...
GET all employees
Method: GET
URL: http://localhost:3000/employees
Send.
Expected response: JSON list of employees.
POST add employee
Method: POST
URL: http://localhost:3000/employees
Body → select raw → JSON
Example JSON:
{
"name": "Charlie Nguyen",
"role": "Developer",
"salary": 70000
}
Send → should return 201 with new
index.js
Describe what to build
Auto
Local
Default Approvals
Go Live


File Edit Selection View Go
Ass15-2
EXPLORER
OPEN EDITORS
index.js
http://localhos...
ASS15-2
node_modules
index.js
package-lock.json
package.json
OUTLINE
TIMELINE
http://localhost:3000/employees/1/salary
Save
No Environment
PUT
http://localhost:3000/employees/1/salary
Send
Params Authorization Headers (9) Body Scripts Settings
none form-data x-www-form-urlencoded raw binary GraphQL
{
"salary": 85000
}
Body Cookies Headers (7) Test Results
Status: 200 OK Time: 14 ms Size: 308 B
Pretty Raw Preview JSON
{"data":[{"id":1,"name":"Alice Johnson","role":"Engineer",
"salary":80000},{"id":2,"name":"Bob Patel","role":"Manager",
"salary":95000}]}
CHAT
NODE.JS EXPRESS REST API FOR EMPLO...
"role": "Developer",
"salary": 70000
}
Send → should return 201 with new employee.
PUT update salary
Method: PUT
URL:
http://localhost:3000/employees/1/salar
Body → raw → JSON
Example:
{
"salary": 85000
}
Send → should return updated employee.
index.js
Describe what to build
Auto
Local
Default Approvals
Go Live



Explanation

This API manages employee records and salary details.
The **GET /employees** endpoint retrieves all employee data.
The **POST /employees** endpoint adds a new employee with salary information.

The **PUT /employees/:id/salary** endpoint updates the salary of a specific employee.
The **DELETE /employees/:id** endpoint removes an employee from the system.
Basic validation ensures correct data input before processing.

Task 3: Student Information REST API

Scenario:

A university wants to digitize student records for easy retrieval. You need to build

a Student Management API that allows:

• Adding a new student (name, roll number, course, year)

• Fetching details of all students

• Updating student information

• Deleting a student record

Implementation Instructions:

1. Use Node.js + Express (or Python Flask/Django REST) with GitHub

Copilot to auto-generate routes.

2. Store student data in an in-memory array or SQLite/PostgreSQL

database.

Test API endpoints with Postman.

Prompt:

Create a Student Management REST API using Node.js and Express.

Features:

- Add student (name, roll number, course, year)

- Get all students

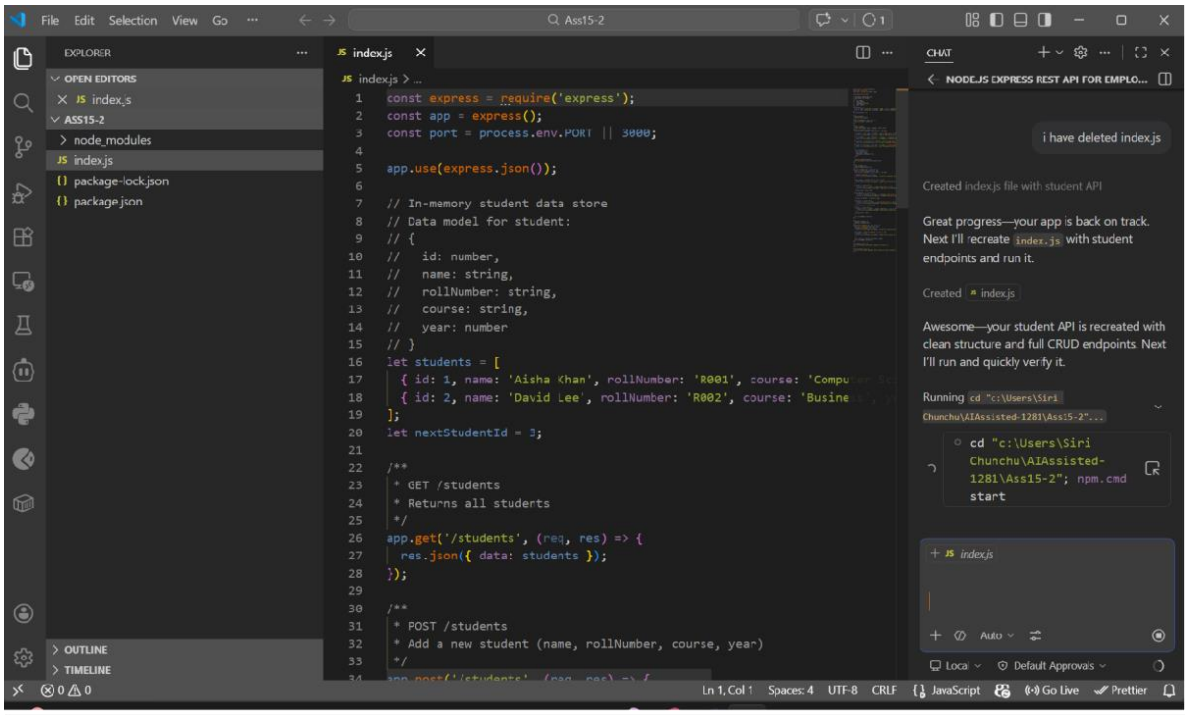
- Update student details

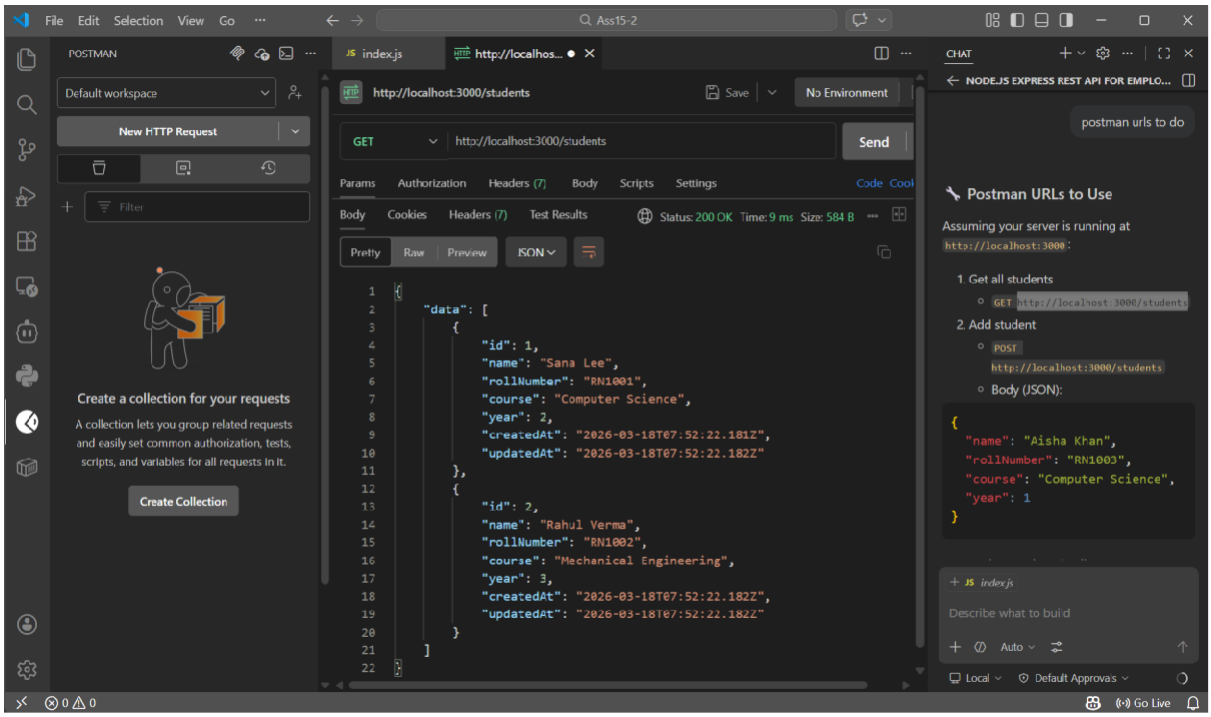
- Delete student

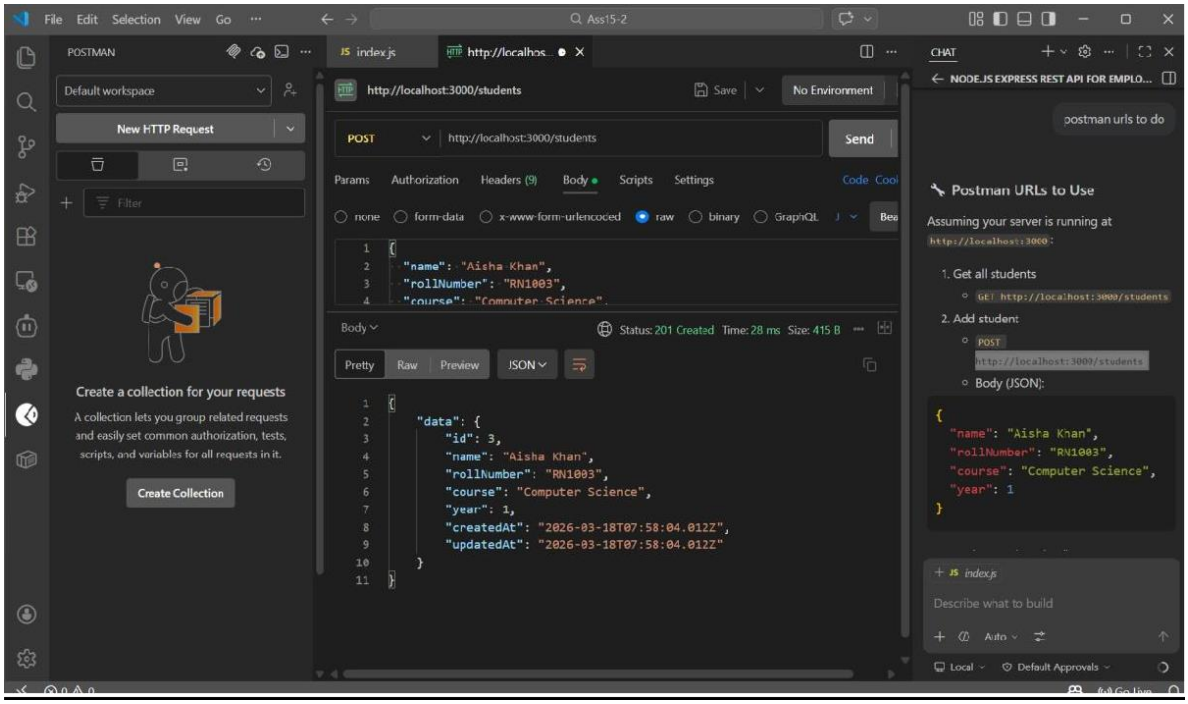
Store data in an in-memory array.

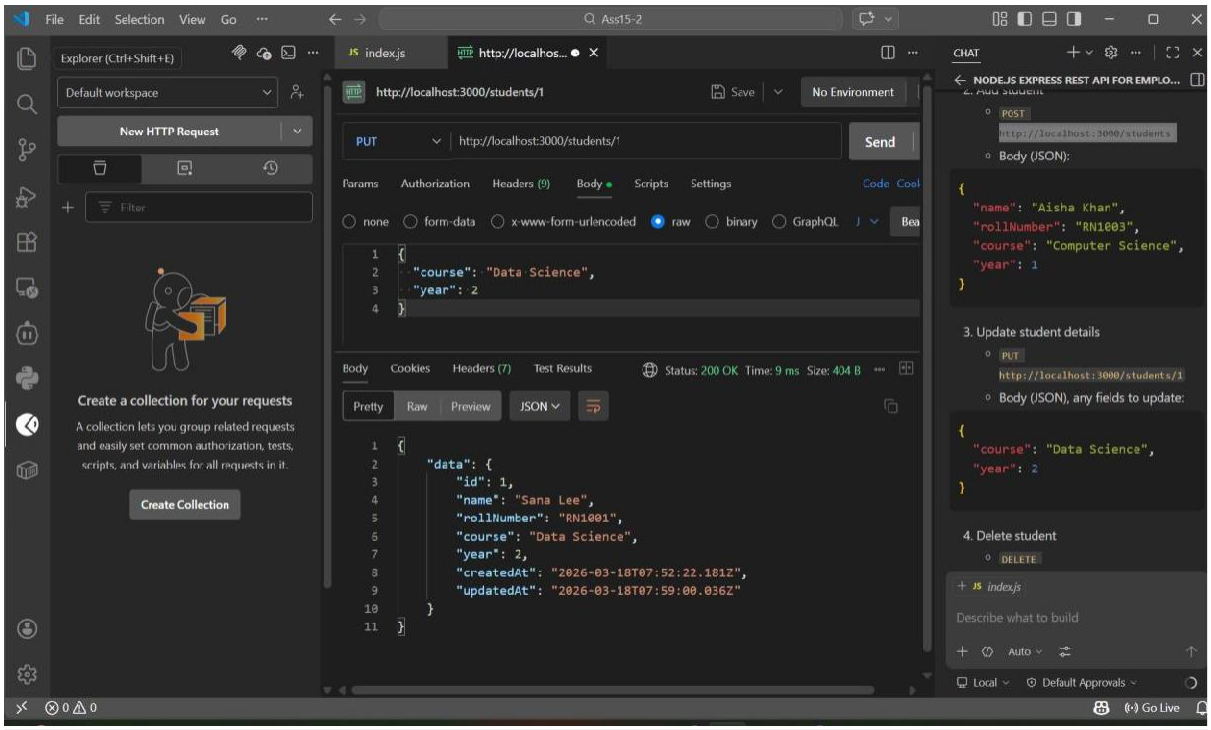
Add comments and clean structure.

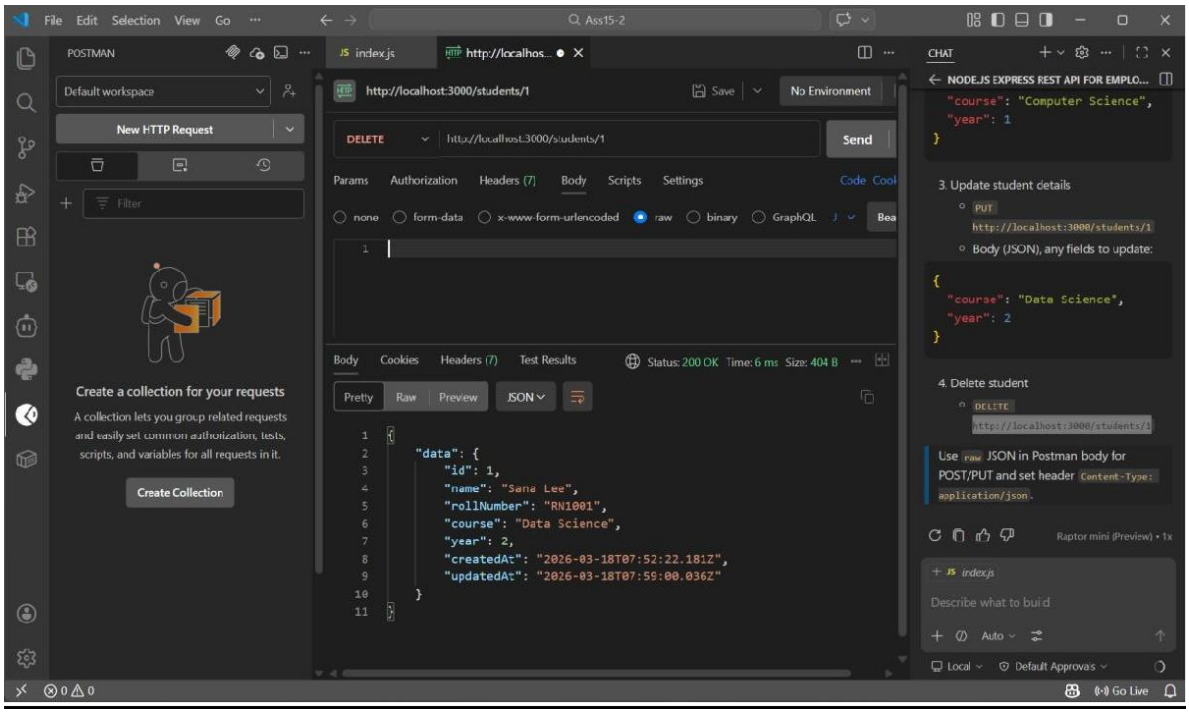
AI Generated:

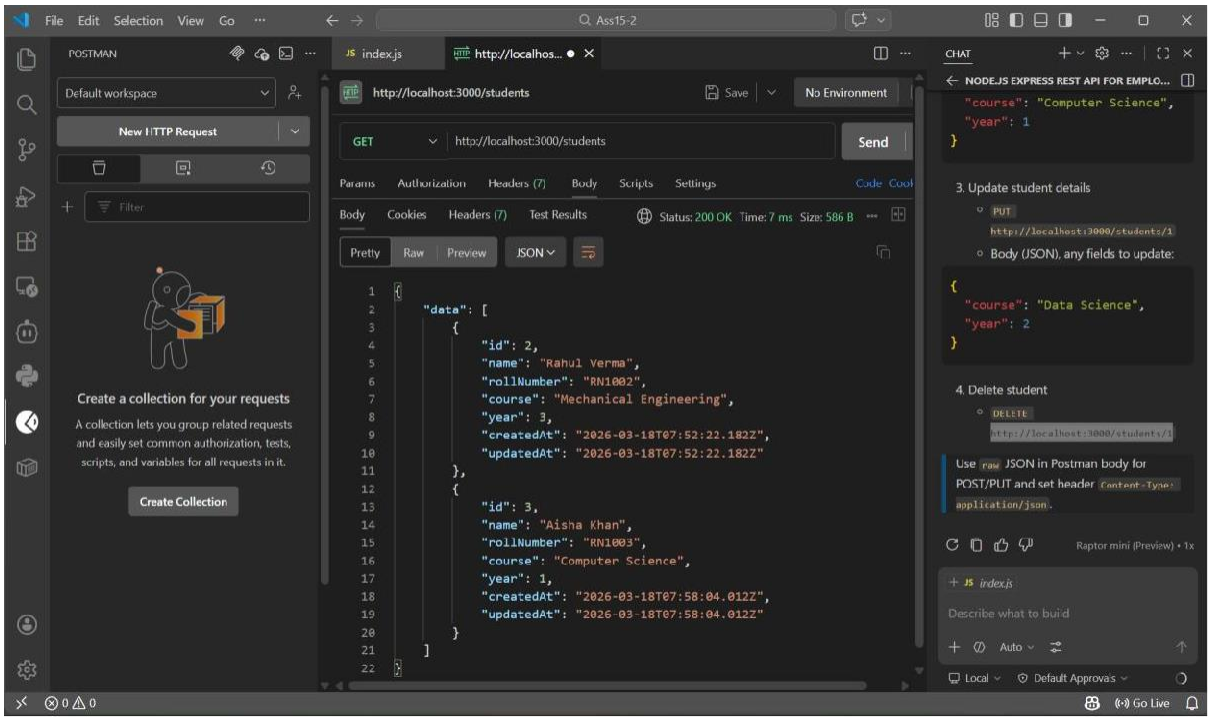



File Edit Selection View Go
Ass15-2
POSTMAN
Default workspace
New HTTP Request
Filter
Create a collection for your requests
A collection lets you group related requests and easily set common authorization, tests, scripts, and variables for all requests in it.
Create Collection
index.js
http://localhos...
http://localhost:3000/students
Save
No Environment
GET
http://localhost:3000/students
Send
Params Authorization Headers (7) Body Scripts Settings
Code Cool
Body Cookies Headers (7) Test Results
Status: 200 OK Time: 9 ms Size: 584 B
Pretty Raw Preview JSON
{
"data": [
{
"id": 1,
"name": "Sana Lee",
"rollNumber": "RN1001",
"course": "Computer Science",
"year": 2,
"createdAt": "2026-03-18T07:52:22.181Z",
"updatedAt": "2026-03-18T07:52:22.182Z"
},
{
"id": 2,
"name": "Rahul Verma",
"rollNumber": "RN1002",
"course": "Mechanical Engineering",
"year": 3,
"createdAt": "2026-03-18T07:52:22.182Z",
"updatedAt": "2026-03-18T07:52:22.182Z"
}
]
}
CHAT
NODE.JS EXPRESS REST API FOR EMPLO...
postman urls to do
Postman URLs to Use
Assuming your server is running at http://localhost:3000:
1. Get all students
GET http://localhost:3000/students
2. Add student
POST http://localhost:3000/students
Body (JSON):
{
"name": "Aisha Khan",
"rollNumber": "RN1003",
"course": "Computer Science",
"year": 1
}
index.js
Describe what to build
Auto
Local
Default Approvals
Go Live


File Edit Selection View Go
Ass15-2
POSTMAN
Default workspace
New HTTP Request
Filter
Create a collection for your requests
A collection lets you group related requests and easily set common authorization, tests, scripts, and variables for all requests in it.
Create Collection
index.js
http://localhos...
http://localhost:3000/students
Save
No Environment
POST
http://localhost:3000/students
Send
Params Authorization Headers (9) Body Scripts Settings
Code Cool
none form-data x-www-form-urlencoded raw binary GraphQL
{
"name": "Aisha Khan",
"rollNumber": "RN1003",
"course": "Computer Science",
Body
Status: 201 Created Time: 28 ms Size: 415 B
Pretty Raw Preview JSON
{
"data": {
"id": 3,
"name": "Aisha Khan",
"rollNumber": "RN1003",
"course": "Computer Science",
"year": 1,
"createdAt": "2026-03-18T07:58:04.012Z",
"updatedAt": "2026-03-18T07:58:04.012Z"
}
}
CHAT
NODE.JS EXPRESS REST API FOR EMPLO...
postman urls to do
Postman URLs to Use
Assuming your server is running at http://localhost:3000:
1. Get all students
GET http://localhost:3000/students
2. Add student
POST http://localhost:3000/students
Body (JSON):
{
"name": "Aisha Khan",
"rollNumber": "RN1003",
"course": "Computer Science",
"year": 1
}
index.js
Describe what to build
Auto
Local
Default Approvals


File Edit Selection View Go
Ass15-2
Explorer (Ctrl+Shift+E)
Default workspace
New HTTP Request
Filter
Create a collection for your requests
A collection lets you group related requests and easily set common authorization, tests, scripts, and variables for all requests in it.
Create Collection
index.js
http://localhos...
http://localhost:3000/students/1
Save
No Environment
PUT
http://localhost:3000/students/1
Send
Params Authorization Headers (9) Body Scripts Settings
none form-data x-www-form-urlencoded raw binary GraphQL
{
"course": "Data Science",
"year": 2
}
Body Cookies Headers (7) Test Results
Status: 200 OK Time: 9 ms Size: 404 B
Pretty Raw Preview JSON
{
"data": {
"id": 1,
"name": "Sana Lee",
"rollNumber": "RN1001",
"course": "Data Science",
"year": 2,
"createdAt": "2026-03-18T07:52:22.181Z",
"updatedAt": "2026-03-18T07:59:00.036Z"
}
}
CHAT
NODEJS EXPRESS REST API FOR EMPLO...
POST
http://localhost:3000/students
Body (JSON):
{
"name": "Aisha Khan",
"rollNumber": "RN1003",
"course": "Computer Science",
"year": 1
}
3. Update student details
PUT
http://localhost:3000/students/1
Body (JSON), any fields to update:
{
"course": "Data Science",
"year": 2
}
4. Delete student
DELETE
index.js
Describe what to build
Auto
Local
Default Approvals
Go Live


File Edit Selection View Go
Ass15-2
POSTMAN
Default workspace
New HTTP Request
Filter
Create a collection for your requests
A collection lets you group related requests and easily set common authorization, tests, scripts, and variables for all requests in it.
Create Collection
index.js
http://localhos...
http://localhost:3000/students/1
Save
No Environment
DELETE
http://localhost:3000/students/1
Send
Params Authorization Headers (7) Body Scripts Settings
none form-data x-www-form-urlencoded raw binary GraphQL
Body Cookies Headers (7) Test Results
Status: 200 OK Time: 6 ms Size: 404 B
Pretty Raw Preview JSON
{
"data": {
"id": 1,
"name": "Sana Lee",
"rollNumber": "RN1001",
"course": "Data Science",
"year": 2,
"createdAt": "2026-03-18T07:52:22.181Z",
"updatedAt": "2026-03-18T07:59:00.036Z"
}
}
CHAT
NODEJS EXPRESS REST API FOR EMPLO...
"course": "Computer Science",
"year": 1
}
3. Update student details
PUT
http://localhost:3000/students/1
Body (JSON), any fields to update:
{
"course": "Data Science",
"year": 2
}
4. Delete student
DELETE
http://localhost:3000/students/1
Use raw JSON in Postman body for POST/PUT and set header Content-Type: application/json.
Raptor mini (Preview) • 1x
index.js
Describe what to build
Auto
Local
Default Approvals
Go Live



Explanation

This API is designed to manage student records in a university system.
The **POST /students** endpoint adds new student details.
The **GET /students** endpoint retrieves all student records.
The **PUT /students/:id** endpoint updates existing student information.
The **DELETE /students/:id** endpoint deletes a student record.
Data is stored in an in-memory array for simplicity.

Task 4: Weather Information API Integration

Scenario:

Students often want to check weather before attending campus activities. Build a backend service that fetches real-time weather data from a public API (e.g., OpenWeatherMap).

Implementation Instructions:

1. Create endpoints like:

• /weather/current/:city → fetches current weather

• /weather/forecast/:city → fetches 5-day forecast

2. Use GitHub Copilot to generate code that calls the external API.

Return results in structured JSON format with temperature, humidity, and condition.

<u>Prompt:</u>

Create a Node.js Express API that integrates with OpenWeatherMap API.

Endpoints:

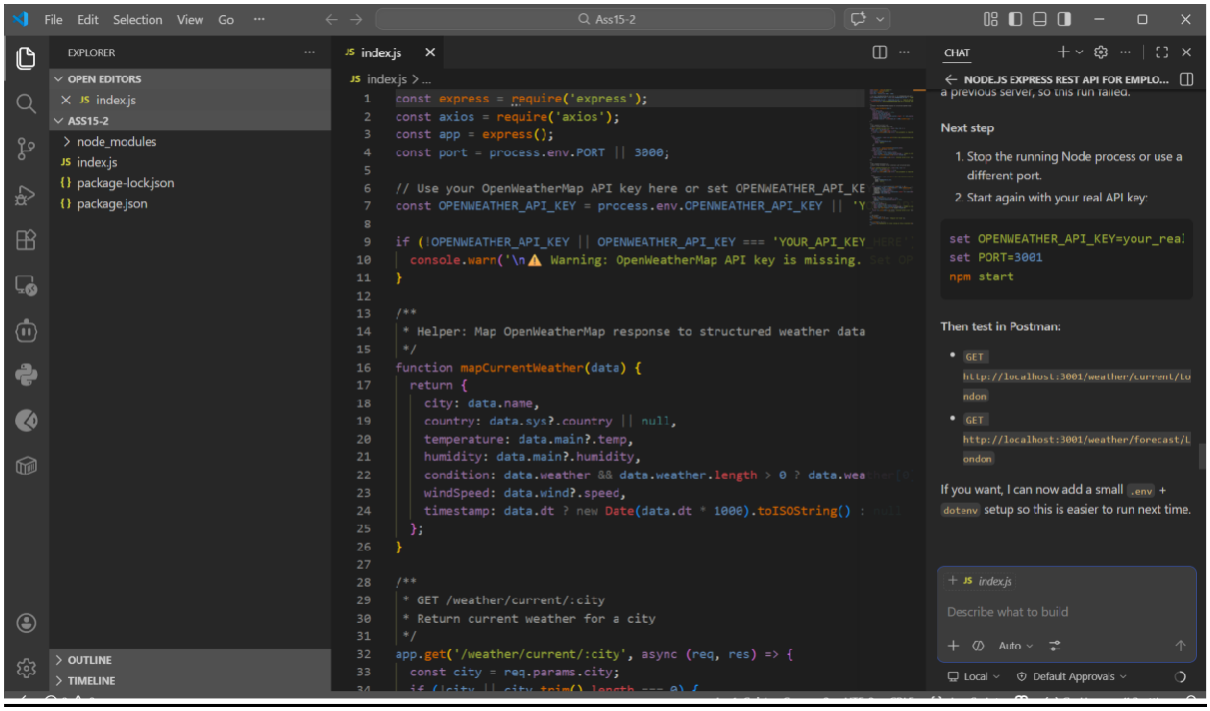
- /weather/current/:city → return temperature, humidity, condition

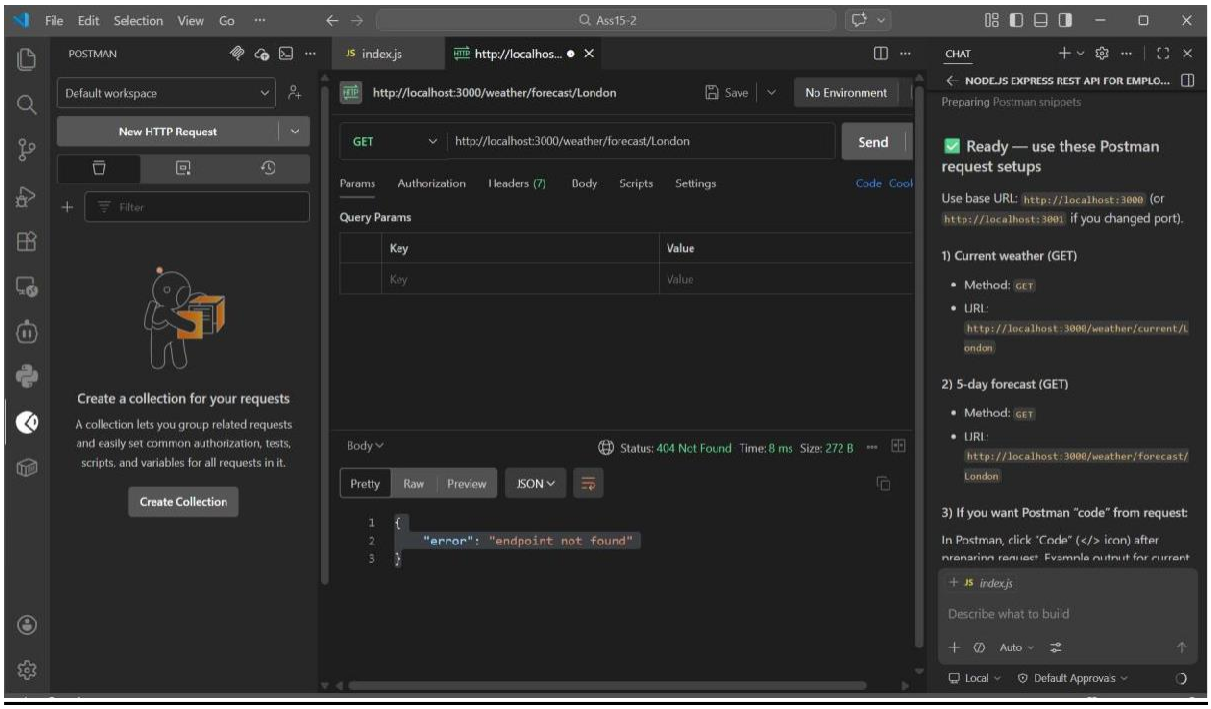
- /weather/forecast/:city → return 5-day forecast

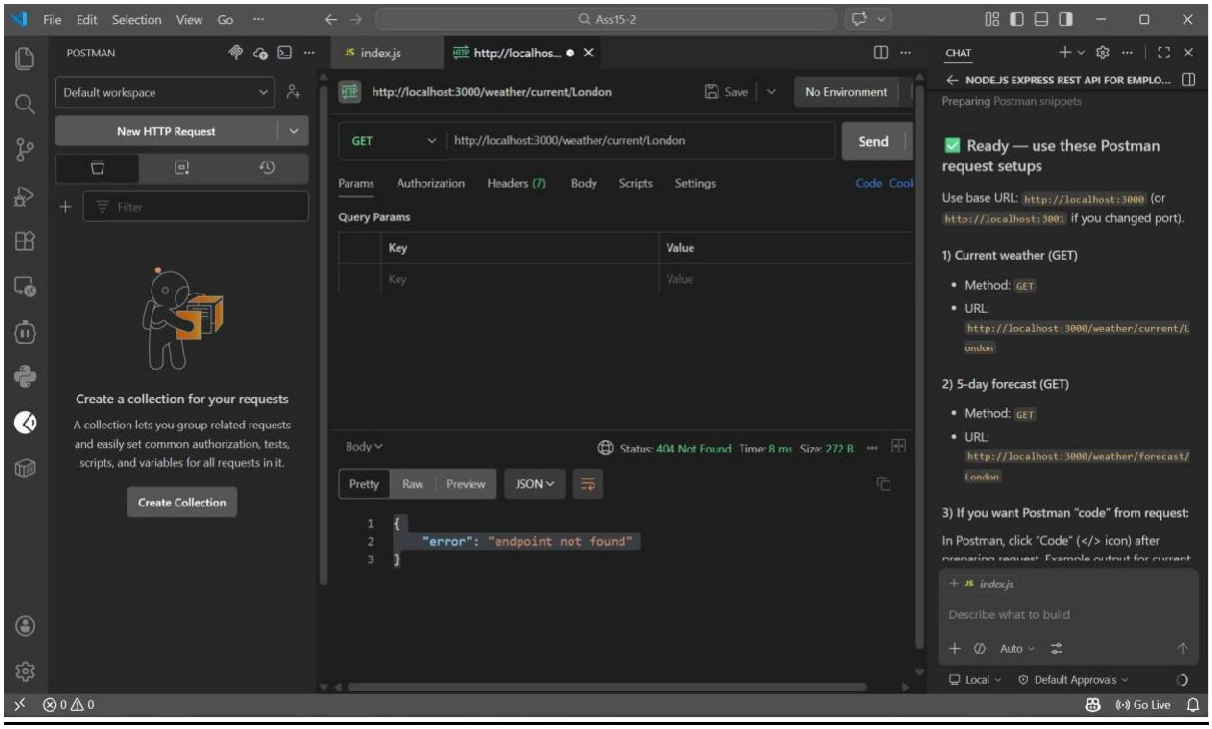
Use axios for API calls and return structured JSON.

Include error handling and comments.

<u>AI Generated:</u>







Explanation

This API integrates with an external service (OpenWeatherMap) to fetch real-time weather data.

The **/weather/current/:city** endpoint retrieves current weather details such as temperature, humidity, and condition.

The **/weather/forecast/:city** endpoint provides forecast data.

Axios is used to make HTTP requests to the external API, and structured JSON responses are returned to the user.

Task 5: Delete Item

Instructions:

• Use AI to create a DELETE endpoint to remove an item by index or ID.

Starter Code:

```
# DELETE /items/<int:index>
@app.route('/items/<int:index>', methods=['DELETE'])
def delete_item(index):
if index < 0 or index >= len(items):
return jsonify({"error": "Item not found"}), 404
removed_item = items.pop(index)
return jsonify({"message": "Item deleted", "item": removed_item})
```

Expected Output:

• DELETE /items/0 →

{"message": "Item deleted", "item": {"name":"Notebook","price":250}}

GET /items → []

Note: Report should be submitted as a word document for all tasks in a single document with prompts, comments & code explanation, and output and if required, screenshots.

<u>Prompt:</u>

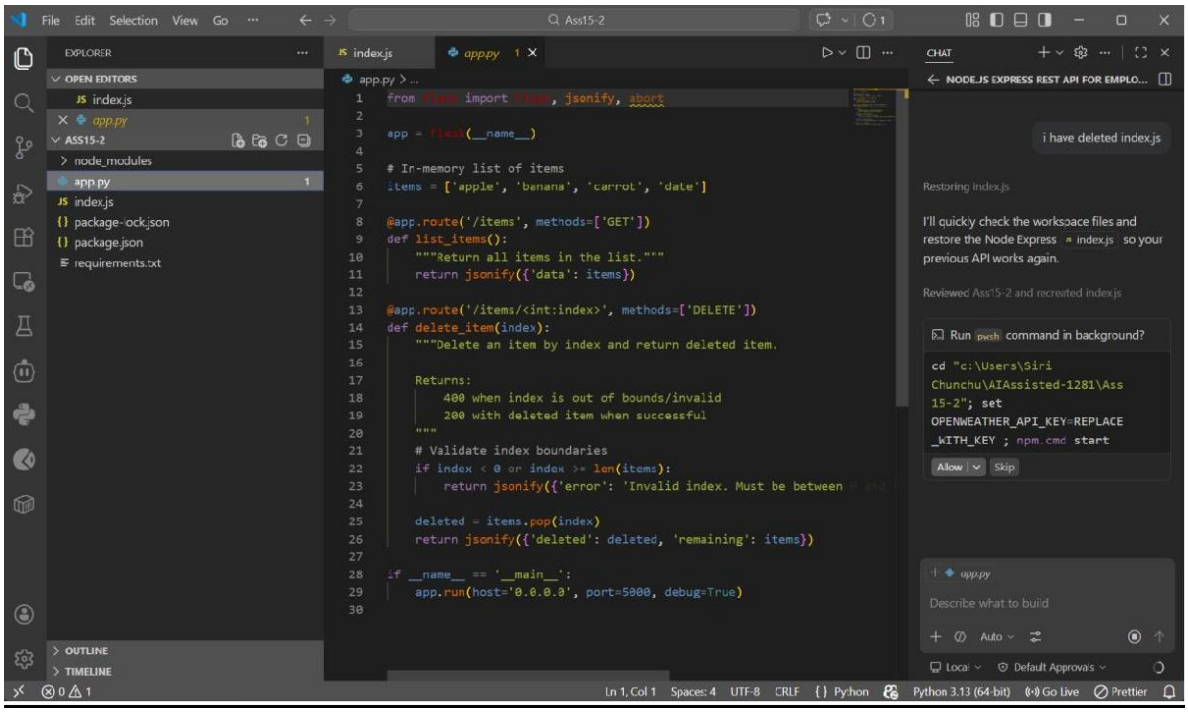
Create a Flask API endpoint to delete an item from a list by index.

Requirements:

- DELETE /items/<int:index>
- Return error if index invalid
- Return deleted item in response

Add proper comments.

<u>AI Generated:</u>


File Edit Selection View Go
Ass15-2
EXPLORER
OPEN EDITORS
index.js
app.py
ASS15-2
node_modules
app.py
index.js
package-lock.json
package.json
requirements.txt
OUTLINE
TIMELINE
index.js
app.py
app.py > ...
from flask import Flask, jsonify, abort
app = Flask(__name__)
# In-memory list of items
items = ['apple', 'banana', 'carrot', 'date']
@app.route('/items', methods=['GET'])
def list_items():
"""Return all items in the list."""
return jsonify({'data': items})
@app.route('/items/<int:index>', methods=['DELETE'])
def delete_item(index):
"""Delete an item by index and return deleted item.
Returns:
400 when index is out of bounds/invalid
200 with deleted item when successful
"""
# Validate index boundaries
if index < 0 or index >= len(items):
return jsonify({'error': 'Invalid index. Must be between
deleted = items.pop(index)
return jsonify({'deleted': deleted, 'remaining': items})
if __name__ == '__main__':
app.run(host='0.0.0.0', port=5000, debug=True)
CHAT
NODE.JS EXPRESS REST API FOR EMPLO...
i have deleted index.js
Restoring index.js
I'll quicky check the workspace files and restore the Node Express index.js so your previous API works again.
Reviewed Ass15-2 and recreated index.js
Run pwsh command in background?
cd "c:\Users\Siri Chunchu\AIAssisted-1281\Ass 15-2"; set OPENWEATHER_API_KEY=REPLACE _WITH_KEY ; npm.cmd start
Allow
Skip
app.py
Describe what to build
Auto
Local
Default Approvals
Ln 1, Col 1
Spaces: 4
UTF-8
CRLF
Python
Python 3.13 (64-bit)
Go Live
Prettier

DELETE http://127.0.0.1:5000/items/0

Params Authorization Headers (2) Body Pre-request Script Tests

none form-data x-www-form-urlencoded binary Graph

```
1 {
2     "message": "Item deleted",
3     "item": {
4         "name": "Notebook",
5         "price": 250
6     }
7 }
```

Body Cookies Preseies Test (1) Test Results

Pretty Raw Preview Visualize JSON

```
1 {
2
```

GET http://127.0.0.1:5000/items

Params Authorization Headers (2) Body Pre-request Script Tests

none form-data x-www-form-urlencoded binary Graph

```
1 }
```

Body Cookies Preseies Test (1) Test Results

Pretty Raw Preview Visualize JSON

Explanation

This API demonstrates how to delete an item from a list using a DELETE request.
The **DELETE /items/<index>** endpoint removes an item based on its index.
It includes error handling to check if the index is valid.
If valid, the item is removed and returned in the response.